ELEMENTARY REAL ANALYSIS

Second Edition (2008)

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This version of **Elementary Real Analysis** contains all the chapters of the text. There are also two smaller editions, **Elementary Real Analysis, Volume One** containing only Chapters 1–8 of the text, and **Elementary Real Analysis, Volume Two** containing Chapters 9–13. The shorter versions are also available as a trade paperbacks. For further information on this title and others in the series visit our website. There are FREE PDF files of all of our texts available for download as well as instructions on how to order trade paperback copies.

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Original Citation: *Elementary Real Analysis*, Brian S. Thomson, Judith B. Bruckner, Andrew M. Bruckner. Prentice-Hall, 2001, xv 735 pp. [ISBN 0-13-019075-61]

Cover Design and Photography: David Sprecher

Second Edition ISBN 143484367X

New Citation: *Elementary Real Analysis, Second Edition*, Brian S. Thomson, Judith B. Bruckner, Andrew M. Bruckner. ClassicalRealAnalysis.com, 2008, xvi 740 pp. [ISBN 143484367X]


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Date PDF file compiled: March 28, 2008

CONTENTS

PREFACE **xii**

VOLUME ONE **1**

1 PROPERTIES OF THE REAL NUMBERS **1**
1.1 Introduction 1
1.2 The Real Number System 2
1.3 Algebraic Structure 4
1.4 Order Structure 7
1.5 Bounds 8
1.6 Sups and Infs 9
1.7 The Archimedean Property 12
1.8 Inductive Property of $\mathbb{N}$ 13
1.9 The Rational Numbers Are Dense 14
1.10 The Metric Structure of $\mathbb{R}$ 16
1.11 Challenging Problems for Chapter 1 18

2 SEQUENCES **21**
2.1 Introduction 21
2.2 Sequences 22
2.2.1 Sequence Examples 23
2.3 Countable Sets 27
2.4 Convergence 29
2.5 Divergence 33
2.6 Boundedness Properties of Limits 35
2.7 Algebra of Limits 37
2.8 Order Properties of Limits 42
2.9 Monotone Convergence Criterion 47
2.10 Examples of Limits 50
2.11 Subsequences 54
2.12 Cauchy Convergence Criterion 58
2.13 Upper and Lower Limits 61
2.14 Challenging Problems for Chapter 2 66

3 INFINITE SUMS **72**
3.1 Introduction 72
3.2 Finite Sums 73
3.3 Infinite Unordered sums 78
3.3.1 Cauchy Criterion 79
3.4 Ordered Sums: Series 83
3.4.1 Properties 84
3.4.2 Special Series 85
3.5 Criteria for Convergence 90
3.5.1 Boundedness Criterion 91
3.5.2 Cauchy Criterion 91
3.5.3 Absolute Convergence 92
3.6 Tests for Convergence 95
3.6.1 Trivial Test 96
3.6.2 Direct Comparison Tests 96
3.6.3 Limit Comparison Tests 98
3.6.4 Ratio Comparison Test 99
3.6.5 d'Alembert's Ratio Test 100
3.6.6 Cauchy's Root Test 101
3.6.7 Cauchy's Condensation Test 103
3.6.8 Integral Test 104
3.6.9 Kummer's Tests 105
3.6.10 Raabe's Ratio Test 107
3.6.11 Gauss's Ratio Test 108
3.6.12 Alternating Series Test 110
3.6.13 Dirichlet's Test 111
3.6.14 Abel's Test 113
3.7 Rearrangements 117
3.7.1 Unconditional Convergence 118
3.7.2 Conditional Convergence 119
3.7.3 Comparison of $\sum_{i=1}^{\infty} a_i$ and $\sum_{i\in\mathbb{N}} a_i$ 120
3.8 Products of Series 123
3.8.1 Products of Absolutely Convergent Series 125
3.8.2 Products of Nonabsolutely Convergent Series 126
3.9 Summability Methods 128
3.9.1 Cesàro's Method 128
3.9.2 Abel's Method 130
3.10 More on Infinite Sums 134
3.11 Infinite Products 135
3.12 Challenging Problems for Chapter 3 139

4 SETS OF REAL NUMBERS **146**
4.1 Introduction 146

4.2 Points 147
4.2.1 Interior Points 147
4.2.2 Isolated Points 148
4.2.3 Points of Accumulation 149
4.2.4 Boundary Points 150
4.3 Sets 153
4.3.1 Closed Sets 153
4.3.2 Open Sets 154
4.4 Elementary Topology 159
4.5 Compactness Arguments 162
4.5.1 Bolzano-Weierstrass Property 163
4.5.2 Cantor's Intersection Property 164
4.5.3 Cousin's Property 166
4.5.4 Heine-Borel Property 168
4.5.5 Compact Sets 171
4.6 Countable Sets 173
4.7 Challenging Problems for Chapter 4 175

5 CONTINUOUS FUNCTIONS 179
5.1 Introduction to Limits 179
5.1.1 Limits (ε-δ Definition) 179
5.1.2 Limits (Sequential Definition) 183
5.1.3 Limits (Mapping Definition) 185
5.1.4 One-Sided Limits 186
5.1.5 Infinite Limits 188
5.2 Properties of Limits 190
5.2.1 Uniqueness of Limits 190
5.2.2 Boundedness of Limits 191
5.2.3 Algebra of Limits 192
5.2.4 Order Properties 195
5.2.5 Composition of Functions 198
5.2.6 Examples 200
5.3 Limits Superior and Inferior 205
5.4 Continuity 207
5.4.1 How to Define Continuity 207
5.4.2 Continuity at a Point 210
5.4.3 Continuity at an Arbitrary Point 213
5.4.4 Continuity on a Set 215
5.5 Properties of Continuous Functions 218
5.6 Uniform Continuity 219
5.7 Extremal Properties 222
5.8 Darboux Property 223
5.9 Points of Discontinuity 225

5.9.1 Types of Discontinuity 225
5.9.2 Monotonic Functions 227
5.9.3 How Many Points of Discontinuity? 230
5.10 Challenging Problems for Chapter 5 232

6 MORE ON CONTINUOUS FUNCTIONS AND SETS 239
6.1 Introduction 239
6.2 Dense Sets 239
6.3 Nowhere Dense Sets 241
6.4 The Baire Category Theorem 243
6.4.1 A Two-Player Game 243
6.4.2 The Baire Category Theorem 244
6.4.3 Uniform Boundedness 246
6.5 Cantor Sets 247
6.5.1 Construction of the Cantor Ternary Set 247
6.5.2 An Arithmetic Construction of K 250
6.5.3 The Cantor Function 252
6.6 Borel Sets 253
6.6.1 Sets of Type G_δ 254
6.6.2 Sets of Type F_σ 256
6.7 Oscillation and Continuity 257
6.7.1 Oscillation of a Function 258
6.7.2 The Set of Continuity Points 260
6.8 Sets of Measure Zero 263
6.9 Challenging Problems for Chapter 6 268

7 DIFFERENTIATION 271
7.1 Introduction 271
7.2 The Derivative 271
7.2.1 Definition of the Derivative 272
7.2.2 Differentiability and Continuity 276
7.2.3 The Derivative as a Magnification 277
7.3 Computations of Derivatives 278
7.3.1 Algebraic Rules 279
7.3.2 The Chain Rule 281
7.3.3 Inverse Functions 285
7.3.4 The Power Rule 286
7.4 Continuity of the Derivative? 288
7.5 Local Extrema 290
7.6 Mean Value Theorem 292
7.6.1 Rolle's Theorem 292
7.6.2 Mean Value Theorem 294
7.6.3 Cauchy's Mean Value Theorem 297

7.7 Monotonicity 298
7.8 Dini Derivates 300
7.9 The Darboux Property of the Derivative 304
7.10 Convexity 307
7.11 L'Hôpital's Rule 312
7.11.1 L'Hôpital's Rule: $\frac{0}{0}$ Form 313
7.11.2 L'Hôpital's Rule as $x \to \infty$ 315
7.11.3 L'Hôpital's Rule: $\frac{\infty}{\infty}$ Form 316
7.12 Taylor Polynomials 319
7.13 Challenging Problems for Chapter 7 322

8 THE INTEGRAL 331
8.1 Introduction 331
8.2 Cauchy's First Method 333
8.2.1 Scope of Cauchy's First Method 335
8.3 Properties of the Integral 338
8.4 Cauchy's Second Method 343
8.5 Cauchy's Second Method (Continued) 345
8.6 The Riemann Integral 347
8.6.1 Some Examples 349
8.6.2 Riemann's Criteria 350
8.6.3 Lebesgue's Criterion 352
8.6.4 What Functions Are Riemann Integrable? 354
8.7 Properties of the Riemann Integral 356
8.8 The Improper Riemann Integral 359
8.9 More on the Fundamental Theorem of Calculus 361
8.10 Challenging Problems for Chapter 8 363

VOLUME TWO 365

9 SEQUENCES AND SERIES OF FUNCTIONS 366
9.1 Introduction 366
9.2 Pointwise Limits 367
9.3 Uniform Limits 373
9.3.1 The Cauchy Criterion 375
9.3.2 Weierstrass M-Test 377
9.3.3 Abel's Test for Uniform Convergence 378
9.4 Uniform Convergence and Continuity 383
9.4.1 Dini's Theorem 385
9.5 Uniform Convergence and the Integral 387
9.5.1 Sequences of Continuous Functions 387
9.5.2 Sequences of Riemann Integrable Functions 389
9.5.3 Sequences of Improper Integrals 391

9.6 Uniform Convergence and Derivatives 393
9.6.1 Limits of Discontinuous Derivatives 394
9.7 Pompeiu's Function 397
9.8 Continuity and Pointwise Limits 399
9.9 Challenging Problems for Chapter 9 402

10 POWER SERIES 404
10.1 Introduction 404
10.2 Power Series: Convergence 404
10.3 Uniform Convergence 410
10.4 Functions Represented by Power Series 412
10.4.1 Continuity of Power Series 412
10.4.2 Integration of Power Series 413
10.4.3 Differentiation of Power Series 414
10.4.4 Power Series Representations 416
10.5 The Taylor Series 418
10.5.1 Representing a Function by a Taylor Series 420
10.5.2 Analytic Functions 422
10.6 Products of Power Series 424
10.6.1 Quotients of Power Series 425
10.7 Composition of Power Series 426
10.8 Trigonometric Series 428
10.8.1 Uniform Convergence of Trigonometric Series 428
10.8.2 Fourier Series 429
10.8.3 Convergence of Fourier Series 430
10.8.4 Weierstrass Approximation Theorem 433

11 THE EUCLIDEAN SPACES $\mathbb{R}^n$ 437
11.1 The Algebraic Structure of $\mathbb{R}^n$ 437
11.2 The Metric Structure of $\mathbb{R}^n$ 439
11.3 Elementary Topology of $\mathbb{R}^n$ 442
11.4 Sequences in $\mathbb{R}^n$ 445
11.5 Functions and Mappings 448
11.5.1 Functions from $\mathbb{R}^n \to \mathbb{R}$ 449
11.5.2 Functions from $\mathbb{R}^n \to \mathbb{R}^m$ 451
11.6 Limits of Functions from $\mathbb{R}^n \to \mathbb{R}^m$ 453
11.6.1 Definition 453
11.6.2 Coordinate-Wise Convergence 455
11.6.3 Algebraic Properties 458
11.7 Continuity of Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ 458
11.8 Compact Sets in $\mathbb{R}^n$ 461
11.9 Continuous Functions on Compact Sets 462
11.10 Additional Remarks 463

12 DIFFERENTIATION ON $\mathbb{R}^n$ **467**
12.1 Introduction 467
12.2 Partial and Directional Derivatives 467
12.2.1 Partial Derivatives 469
12.2.2 Directional Derivatives 471
12.2.3 Cross Partials 473
12.3 Integrals Depending on a Parameter 476
12.4 Differentiable Functions 480
12.4.1 Approximation by Linear Functions 481
12.4.2 Definition of Differentiability 481
12.4.3 Differentiability and Continuity 485
12.4.4 Directional Derivatives 486
12.4.5 An Example 488
12.4.6 Sufficient Conditions for Differentiability 490
12.4.7 The Differential 492
12.5 Chain Rules 494
12.5.1 Preliminary Discussion 494
12.5.2 Informal Proof of a Chain Rule 497
12.5.3 Notation of Chain Rules 498
12.5.4 Proofs of Chain Rules (I) 500
12.5.5 Mean Value Theorem 502
12.5.6 Proofs of Chain Rules (II) 503
12.5.7 Higher Derivatives 505
12.6 Implicit Function Theorems 507
12.6.1 One-Variable Case 508
12.6.2 Several-Variable Case 511
12.6.3 Simultaneous Equations 514
12.6.4 Inverse Function Theorem 517
12.7 Functions From $\mathbb{R} \to \mathbb{R}^m$ 520
12.8 Functions From $\mathbb{R}^n \to \mathbb{R}^m$ 523
12.8.1 Review of Differentials and Derivatives 524
12.8.2 Definition of the Derivative 526
12.8.3 Jacobians 527
12.8.4 Chain Rules 530
12.8.5 Proof of Chain Rule 532

13 METRIC SPACES **537**
13.1 Introduction 537
13.2 Metric Spaces—Specific Examples 539
13.3 Additional Examples 543
13.3.1 Sequence Spaces 543
13.3.2 Function Spaces 545
13.4 Convergence 548

13.5 Sets in a Metric Space 552
13.6 Functions 558
13.6.1 Continuity 560
13.6.2 Homeomorphisms 564
13.6.3 Isometries 570
13.7 Separable Spaces 573
13.8 Complete Spaces 575
13.8.1 Completeness Proofs 576
13.8.2 Subspaces of a Complete Space 578
13.8.3 Cantor Intersection Property 578
13.8.4 Completion of a Metric Space 579
13.9 Contraction Maps 581
13.10 Applications of Contraction Maps (I) 588
13.11 Applications of Contraction Maps (II) 591
13.11.1 Systems of Equations (Example 13.79 Revisited) 591
13.11.2 Infinite Systems (Example 13.80 revisited) 592
13.11.3 Integral Equations (Example 13.81 revisited) 594
13.11.4 Picard's Theorem (Example 13.82 revisited) 595
13.12 Compactness 597
13.12.1 The Bolzano-Weierstrass Property 597
13.12.2 Continuous Functions on Compact Sets 600
13.12.3 The Heine-Borel Property 602
13.12.4 Total Boundedness 604
13.12.5 Compact Sets in $\mathcal{C}[a,b]$ 607
13.12.6 Peano's Theorem 611
13.13 Baire Category Theorem 614
13.13.1 Nowhere Dense Sets 614
13.13.2 The Baire Category Theorem 617
13.14 Applications of the Baire Category Theorem 620
13.14.1 Functions Whose Graphs "Cross No Lines" 620
13.14.2 Nowhere Monotonic Functions 623
13.14.3 Continuous Nowhere Differentiable Functions 624
13.14.4 Cantor Sets 625
13.15 Challenging Problems for Chapter 13 627

VOLUME ONE **A-1**

A APPENDIX: BACKGROUND **A-1**
A.1 Should I Read This Chapter? A-1
A.2 Notation A-1
A.2.1 Set Notation A-1
A.2.2 Function Notation A-4
A.3 What Is Analysis? A-9

A.4 Why Proofs? A-10
A.5 Indirect Proof A-11
A.6 Contraposition A-12
A.7 Counterexamples A-13
A.8 Induction A-14
A.9 Quantifiers A-17

SUBJECT INDEX **A-20**

PREFACE

Preface to the second edition (January 2008)

This edition differs from the original 2001 version only in that we corrected a number of misprints and other errors. We are grateful to the many users of that version for notifying us of errors they found. We would like to make special mention of Richard Delaware (University of Missouri-Kansas City), and Steve Agronsky (California State Polytechnic University, San Luis Obispo), both of whom went through the entire first edition, made many helpful suggestions, and found numerous errors.

Original Preface (2001)

University mathematics departments have for many years offered courses with titles such as *Advanced Calculus* or *Introductory Real Analysis*. These courses are taken by a variety of students, serve a number of purposes, and are written at various levels of sophistication. The students range from ones who have just completed a course in elementary calculus to beginning graduate students in mathematics. The purposes are multifold:

1. To present familiar concepts from calculus at a more rigorous level.

2. To introduce concepts that are not studied in elementary calculus but that are needed in more advanced undergraduate courses. This would include such topics as point set theory, uniform continuity of functions, and uniform convergence of sequences of functions.

3. To provide students with a level of mathematical sophistication that will prepare them for graduate work in mathematical analysis, or for graduate work in several applied fields such as engineering or economics.

4. To develop many of the topics that the authors feel all students of mathematics should know.

There are now many texts that address some or all of these objectives. These books range from ones that do little more than address objective (1) to ones that try to address all four objectives. The books of the first extreme are generally aimed at one-term courses for students with minimal background. Books at the other extreme often contain substantially more material than can be covered in a one-year course.

The level of rigor varies considerably from one book to another, as does the style of presentation. Some books endeavor to give a very efficient streamlined development; others try to be more user friendly. We have opted for the user-friendly approach. We feel this approach makes the concepts more meaningful to the student.

Our experience with students at various levels has shown that most students have difficulties when topics that are entirely new to them first appear. For some students that might occur almost immediately when rigorous proofs are required, for example, ones needing ε-δ arguments. For others, the difficulties begin with elementary point set theory, compactness arguments, and the like.

To help students with the transition from elementary calculus to a more rigorous course, we have included motivation for concepts most students have not seen before and provided more details in proofs when we introduce new methods. In addition, we have tried to give students ample opportunity to see the new tools in action.

For example, students often feel uneasy when they first encounter the various compactness arguments (Heine-Borel theorem, Bolzano-Weierstrass theorem, Cousin's lemma, introduced in Section 4.5). To help the student see why such theorems are useful, we pose the problem of determining circumstances under which local boundedness of a function f on a set E implies global boundedness of f on E. We show by example that some conditions on E are needed, namely that E be closed and bounded, and then show how each of several theorems could be used to show that closed and boundedness of the set E suffices. Thus we introduce students to the theorems by showing how the theorems can be used in natural ways to solve a problem.

We have also included some optional material, marked as "Advanced" or "Enrichment" and flagged with the symbol ✂.

Enrichment

We have indicated as "Enrichment"' some relatively elementary material that could be added to a longer course to provide enrichment and additional examples. For example, in Chapter 3 we have added to the study of series a section on infinite products. While such a topic plays an important role in the representation of analytic functions, it is presented here to allow the instructor to explore ideas that are closely related to the study of series and that help illustrate and review many of the fundamental ideas that have played a role in the study of series.

Advanced

We have indicated as "Advanced" material of a more mathematically sophisticated nature that can be omitted without loss of continuity. These topics might be needed in more advanced courses in real analysis or in certain of the marked sections or exercises that appear later in this book. For example, in Chapter 2 we have added to the study of sequence limits a section on lim sups and lim infs. For an elementary first course this can be considered somewhat advanced and skipped. Later problems and text material that require these concepts are carefully indicated. Thus, even though the text carries on to relatively advanced undergraduate analysis, a first course can be presented by avoiding these advanced sections.

We apply these markings to some entire chapters as well as to some sections within

chapters and even to certain exercises. We do not view these markings as absolute. They can simply be interpreted in the following ways. Any unmarked material will not depend, in any substantial way, on earlier marked sections. In addition, if a section has been flagged and will be used in a much later section of this book, we indicate where it will be required.

The material marked "Advanced" is in line with goals (2) and (3). We resist the temptation to address objective (4). There are simply too many additional topics that one might feel every student should know (e.g., functions of bounded variation, Riemann-Stieltjes and Lebesgue integrals). To cover these topics in the manner we cover other material would render the book more like a reference book than a text that could reasonably be covered in a year. Students who have completed this book will be in a good position to study such topics at rigorous levels.

We include, however, a chapter on metric spaces. We do this for two reasons: to offer a more general framework for viewing concepts treated in earlier chapters, and to illustrate how the abstract viewpoint can be applied to solving concrete problems. The metric space presentation in Chapter 13 can be considered more advanced as the reader would require a reasonable level of preparation. Even so, it is more readable and accessible than many other presentations of metric space theory, as we have prepared it with the assumption that the student has just the minimal background. For example, it is easier than the corresponding chapter in our graduate level text (*Real Analysis*, Prentice Hall, 1997) in which the student is expected to have studied the Lebesgue integral and to be at an appropriately sophisticated level.

The Exercises

The exercises form an integral part of the book. Many of these exercises are routine in nature. Others are more demanding. A few provide examples that are not usually presented in books of this type but that students have found challenging, interesting, and instructive.

Some exercises have been flagged with the ✂ symbol to indicate that they require material from a flagged section. For example, a first course is likely to skip over the section on lim sups and lim infs of sequences. Exercises that require those concepts are flagged so that the instructor can decide whether they can be used or not. Generally, that symbol on an exercise warns that it might not be suitable for routine assignments.

The exercises at the end of some of the chapters can be considered more challenging. They include some Putnam problems and some problems from the journal *American Mathematical Monthly*. They do not require more knowledge than is in the text material but often need a bit more persistence and some clever ideas. Students should be made aware that solutions to Putnam problems can be found on various Web sites and that solutions to *Monthly* problems are published; even so, the fun in such problems is in the attempt rather than in seeing someone else's solution.

Designing a Course

We have attempted to write this book in a manner sufficiently flexible to make it possible to use the book for courses of various lengths and a variety of levels of mathematical sophistication.

Much of the material in the book involves rigorous development of topics of a relatively elementary nature, topics that most students have studied at a nonrigorous level in a calculus course. A short course of moderate mathematical sophistication intended for students of minimal background can be based entirely on this material. Such a course might meet objective (1).

We have written this book in a leisurely style. This allows us to provide motivational discussions and historical perspective in a number of places. Even though the book is relatively large (in terms of number of pages), we can comfortably cover most of the main sections in a full-year course, including many of the interesting exercises.

Instructors teaching a short course have several options. They can base a course entirely on the unmarked material of Chapters 1, 2, 4, 5, and 7. As time permits, they can add the early parts of Chapters 3 and 8 or parts of Chapters 11 and 12 and some of the enrichment material.

Background

We should make one more point about this book. We do assume that students are familiar with nonrigorous calculus. In particular, we assume familiarity with the elementary functions and their elementary properties. We also assume some familiarity with computing derivatives and integrals. This allows us to illustrate various concepts using examples familiar to the students. For example, we begin Chapter 2, on sequences, with a discussion of approximating $\sqrt{2}$ using Newton's method. This is merely a motivational discussion, so we are not bothered by the fact that we don't treat the derivative formally until Chapter 7 and haven't yet proved that $\frac{d}{dx}(x^2 - 2) = 2x$. For students with minimal background we provide an appendix that informally covers such topics as notation, elementary set theory, functions, and proofs.

Acknowledgments

A number of friends, colleagues, and students have made helpful comments and suggestions while the text was being written. We are grateful to the reviewers of the text: Professors Eugene Allgower (Colorado State University), Stephen Breen (California State University, Northridge), Robert E. Fennell (Clemson University), Jan E. Kucera (Washington State University), and Robert F. Lax (Louisiana State University). The authors are particularly grateful to Professors Steve Agronsky (California Polytechnic State University), Peter Borwein (Simon Fraser University), Paul Humke (St. Olaf College), T. H. Steele (Weber State University), and Clifford Weil (Michigan State University) for using preliminary versions of the book in their classes.

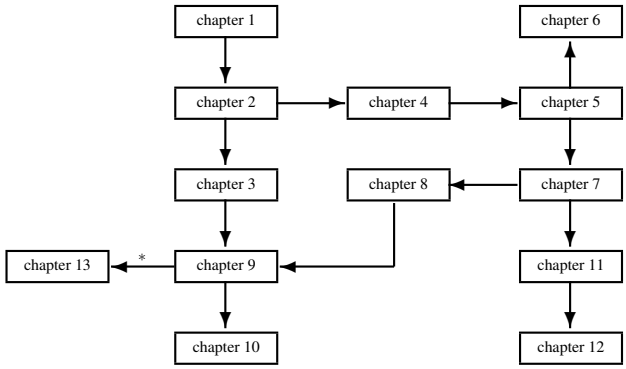


Figure 0.1. Chapter Dependencies (Unmarked Sections).

Chapter 1

PROPERTIES OF THE REAL NUMBERS

Often I have considered the fact that most difficulties which block the progress of students trying to learn analysis stem from this: that although they understand little of ordinary algebra, still they attempt this more subtle art. From this it follows not only that they remain on the fringes, but in addition they entertain strange ideas about the concept of the infinite, which they must try to use. — Leonhard Euler (1748).

1.1 Introduction

The goal of any analysis course is to do some analysis. There are some wonderfully important and interesting facts that can be established in a first analysis course.

Unfortunately, all of the material we wish to cover rests on some foundations, foundations that may not have been properly set down in your earlier courses. Calculus courses traditionally avoid any foundational problems by simply not proving the statements that would need them. Here we cannot do this. We must start with the real number system.

Historically much of real analysis was undertaken without any clear understanding of the real numbers. To be sure the mathematicians of the time had a firm intuitive grasp of the nature of the real numbers and often found precisely the right tool to use in their proofs, but in many cases the tools could not be justified by any line of reasoning.

By the 1870s mathematicians such as Georg Cantor (1845–1918) and Richard Dedekind (1831–1916) had found ways to describe the real numbers that seemed rigorous. We could follow their example and find a presentation of the real numbers that starts at the very beginning and leads up slowly (very slowly) to the exact tools that we need to study analysis. This subject is, perhaps, best left to courses in logic, where other important foundation issues can be discussed.

The approach we shall take (and most textbooks take) is simply to list all the tools that are needed in such a study and take them for granted. You may consider that the real number system is exactly as you have always imagined it. You can sketch pictures of the real line and measure distances and consider the order just as before. Nothing is changed from high school algebra or calculus. But when we come to prove assertions about real numbers or real functions or real sets, we must use exactly the tools here and

not rely on our intuition.

1.2 The Real Number System

To do real analysis we should know exactly what the real numbers are. Here is a loose exposition, suitable for calculus students but (as we will see) not suitable for us.

The Natural Numbers We start with the natural numbers. These are the counting numbers

$$1,2,3,4,\ldots.$$

The symbol $\mathbb{N}$ is used to indicate this collection. Thus $n \in \mathbb{N}$ means that n is a natural number, one of these numbers $1,2,3,4,\ldots$.

There are two operations on the natural numbers, addition and multiplication:

$$m+n \text{ and } m\cdot n.$$

There is also an order relation

$$m < n.$$

Large amounts of time in elementary school are devoted to an understanding of these operations and the order relation.

(Subtraction and division can also be defined, but not for all pairs in $\mathbb{N}$. While $7-5$ and $10/5$ are assigned a meaning [we say $x = 7-5$ if $x+5=7$ and we say $x = 10/5$ if $5\cdot x = 10$] there is no meaning that can be attached to $5-7$ and $5/10$ in this number system.)

The Integers For various reasons, usually well motivated in the lower grades, the natural numbers prove to be rather limited in representing problems that arise in applications of mathematics to the real world. Thus they are enlarged by adjoining the negative integers and zero. Thus the collection

$$\ldots,-4,-3,-2,-1,0,1,2,3,4,\ldots$$

is denoted $\mathbb{Z}$ and called the integers. (The symbol $\mathbb{N}$ seems obvious enough [N for "natural"] but the symbol $\mathbb{Z}$ for the integers originates in the German word for whole number.)

Once again, there are two operations on $\mathbb{Z}$, addition and multiplication:

$$m+n \text{ and } m\cdot n.$$

Again there is an order relation

$$m < n.$$

Fortunately, the rules of arithmetic and order learned for the simpler system $\mathbb{N}$ continue to hold for $\mathbb{Z}$, and young students extend their abilities perhaps painlessly.

(Subtraction can now be defined in this larger number system, but division still may not be defined. For example, $-9/3$ is defined but $3/(-9)$ is not.)

The Rational Numbers At some point the problem of the failure of division in the sets $\mathbb{N}$ and $\mathbb{Z}$ becomes acute and the student must progress to an understanding of fractions. This larger number system is denoted $\mathbb{Q}$, where the symbol chosen is meant to suggest quotients, which is after all what fractions are.

The collection of all "numbers" of the form

$$\frac{m}{n},$$

where $m \in \mathbb{Z}$ and $n \in \mathbb{N}$ is called the set of *rational numbers* and is denoted $\mathbb{Q}$.

A higher level of sophistication is demanded at this stage. Equality has a new meaning. In $\mathbb{N}$ or $\mathbb{Z}$ a statement $m = n$ meant merely that m and n were the same object. Now

$$\frac{m}{n} = \frac{a}{b}$$

for $m, a \in \mathbb{Z}$ and $n, b \in \mathbb{N}$ means that

$$m \cdot b = a \cdot n.$$

Addition and multiplication present major challenges too. Ultimately the students must learn that

$$\frac{m}{n} + \frac{a}{b} = \frac{mb + na}{nb}$$

and

$$\frac{m}{n} \cdot \frac{a}{b} = \frac{ma}{nb}.$$

Subtraction and division are similarly defined. Fortunately, once again the rules of arithmetic are unchanged. The associative rule, distributive rule, etc. all remain true even in this number system.

Again, too, an order relation

$$\frac{m}{n} < \frac{a}{b}$$

is available. It can be defined by requiring, for $m, a \in \mathbb{Z}$ and $n, b \in \mathbb{N}$,

$$mb < na.$$

The same rules for inequalities learned for integers and natural numbers are valid for rationals.

The Real Numbers Up to this point in developing the real numbers we have encountered only arithmetic operations. The progression from $\mathbb{N}$ to $\mathbb{Z}$ to $\mathbb{Q}$ is simply algebraic. All this algebra might have been a burden to the weaker students in the lower grades, but conceptually the steps are easy to grasp with a bit of familiarity.

The next step, needed for all calculus students, is to develop the still larger system of real numbers, denoted as $\mathbb{R}$. We often refer to the real number system as *the real line* and think about it as a geometrical object, even though nothing in our definitions would seem at first sight to allow this.

Most calculus students would be hard pressed to say exactly what these numbers are. They recognize that $\mathbb{R}$ includes all of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ and also many new numbers,

such as $\sqrt{2}$, e, and π. But asked what a real number is, many would return a blank stare. Even just asked what $\sqrt{2}$, e, or π are often produces puzzlement. Well, $\sqrt{2}$ is a number whose square is 2. But is there a number whose square is 2? A calculator might oblige with 1.4142136, but

$$(1.4142136)^2 \neq 2.$$

So what exactly "is" this number $\sqrt{2}$? If we are unable to write down a number whose square is 2, why can we claim that there is a number whose square is 2? And π and e are worse.

Some calculus texts handle this by proclaiming that real numbers are obtained by infinite decimal expansions. Thus while rational numbers have infinite decimal expansions that terminate (e.g., $1/4 = 0.25$) or repeat (e.g., $1/3 = 0.333333\ldots$), the collection of real numbers would include *all* infinite decimal expansions whether repeating, terminating, or not. In that case the claim would be that there is some infinite decimal expansion $1.414213\ldots$ whose square really is 2 and that infinite decimal expansion is the number we mean by the symbol $\sqrt{2}$.

This approach is adequate for applications of calculus and is a useful way to avoid doing any hard mathematics in introductory calculus courses. But you should recall that, at certain stages in the calculus textbook that you used, appeared a phrase such as "the proof of this next theorem is beyond the level of this text." It was beyond the level of the text only because the real numbers had not been properly treated and so there was no way that a proof could have been attempted.

We need to construct such proofs and so we need to abandon this loose, descriptive way of thinking about the real numbers. Instead we will define the real numbers to be a complete, ordered field. In the next sections each of these terms is defined.

1.3 Algebraic Structure

We describe the real numbers by assuming that they have a collection of properties. We do not construct the real numbers, we just announce what properties they are to have. Since the properties that we develop are familiar and acceptable and do in fact describe the real numbers that we are accustomed to using, this approach should not cause any distress. We are just stating rather clearly what it is about the real numbers that we need to use.

We begin with the algebraic structure.

In elementary algebra courses one learns many formulas that are valid for real numbers. For example, the formula

$$(x+y)+z = x+(y+z)$$

called the associative rule is learned. So also is the useful factoring rule

$$x^2 - y^2 = (x-y)(x+y).$$

It is possible to reduce the many rules to one small set of rules that can be used to prove all the other rules.

These rules can be used for other kinds of algebra, algebras where the objects are not real numbers but some other kind of mathematical constructions. This particular structure occurs so frequently, in fact, and in so many different applications that it has its own name. Any set of objects that has these same features is called a *field*. Thus we can say that the first important structure of the real number system is the field structure.

The following nine properties are called the *field axioms*. When we are performing algebraic manipulations in the real number system it is the field axioms that we are really using.

Assume that the set of real numbers $\mathbb{R}$ has two operations, called addition "$+$" and multiplication "$\cdot$" and that these operations satisfy the field axioms. The operation $a \cdot b$ (multiplication) is most often written without the dot as ab.

A1 For any $a, b \in \mathbb{R}$ there is a number $a+b \in \mathbb{R}$ and $a+b = b+a$.

A2 For any $a, b, c \in \mathbb{R}$ the identity

$$(a+b)+c = a+(b+c)$$

is true.

A3 There is a unique number $0 \in \mathbb{R}$ so that, for all $a \in \mathbb{R}$,

$$a+0 = 0+a = a.$$

A4 For any number $a \in \mathbb{R}$ there is a corresponding number denoted by $-a$ with the property that

$$a+(-a) = 0.$$

M1 For any $a, b \in \mathbb{R}$ there is a number $ab \in \mathbb{R}$ and $ab = ba$.

M2 For any $a, b, c \in \mathbb{R}$ the identity

$$(ab)c = a(bc)$$

is true.

M3 There is a unique number $1 \in \mathbb{R}$ so that

$$a1 = 1a = a$$

for all $a \in \mathbb{R}$.

M4 For any number $a \in \mathbb{R}$, $a \neq 0$, there is a corresponding number denoted a^{-1} with the property that

$$aa^{-1} = 1.$$

AM1 For any $a, b, c \in \mathbb{R}$ the identity

$$(a+b)c = ac+bc$$

is true.

Note that we have labeled the axioms with letters indicating which operations are affected, thus A for addition and M for multiplication. The distributive rule AM1 connects addition and multiplication.

How are we to use these axioms? The answer likely is that, in an analysis course, you would not. You might try some of the exercises to understand what a field is and why the real numbers form a field. In an algebra course it would be most interesting to consider many other examples of fields and some of their applications. For an analysis course, understand that we are trying to specify exactly what we mean by the real number system, and these axioms are just the beginning of that process. The first step in that process is to declare that the real numbers form a field under the two operations of addition and multiplication.

Exercises

1.3.1 The field axioms include rules known often as associative rules, commutative rules and distributive rules. Which are which and why do they have these names?

1.3.2 To be precise we would have to say what is meant by the operations of addition and multiplication. Let S be a set and let $S \times S$ be the set of all ordered pairs (s_1, s_2) for s_1, $s_2 \in S$. A *binary operation* on S is a function $B : S \times S \to S$. Thus the operation takes the pair (s_1, s_2) and outputs the element $B(s_1, s_2)$. For example, addition is a binary operation. We could write $(s_1, s_2) \to A(s_1, s_2)$ rather than the more familiar $(s_1, s_2) \to s_1 + s_2$.

(a) Rewrite axioms A1–A4 using this notation $A(s_1, s_2)$ instead of the sum notation.

(b) Define a binary operation on $\mathbb{R}$ different from addition, subtraction, multiplication, or division and determine some of its properties.

(c) For a binary operation B define what you might mean by the commutative, associative, and distributive rules.

(d) Does the binary operation of subtraction satisfy any one of the commutative, associative, or distributive rules?

1.3.3 If in the field axioms for $\mathbb{R}$ we replace $\mathbb{R}$ by any other set with two operations $+$ and $\cdot$ that satisfy these nine properties, then we say that that structure is a field. For example, $\mathbb{Q}$ is a field. The rules are valid since $\mathbb{Q} \subset \mathbb{R}$. The only thing that needs to be checked is that $a + b$ and $a \cdot b$ are in $\mathbb{Q}$ if both a and b are. For this reason $\mathbb{Q}$ is called a *subfield* of $\mathbb{R}$. Find another subfield.

See Note 1

1.3.4 Let S be a set consisting of two elements labeled as A and B. Define $A + A = A$, $B + B = A$, $A + B = B + A = B$, $A \cdot A = A$, $A \cdot B = B \cdot A = A$, and $B \cdot B = B$. Show that all nine of the axioms of a field hold for this structure.

1.3.5 Using just the field axioms, show that

$$(x+1)^2 = x^2 + 2x + 1$$

for all $x \in \mathbb{R}$. Would this identity be true in any field?

See Note 2

1.3.6 Define operations of addition and multiplication on $\mathbb{Z}_5 = \{0,1,2,3,4\}$ as follows:

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Show that $\mathbb{Z}_5$ satisfies all the field axioms.

1.3.7 Define operations of addition and multiplication on $\mathbb{Z}_6 = \{0,1,2,3,4,5\}$ as follows:

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Which of the field axioms does $\mathbb{Z}_6$ fail to satisfy?

1.4 Order Structure

The real number system also enjoys an order structure. Part of our usual picture of the reals is the sense that some numbers are "bigger" than others or more to the "right" than others. We express this by using inequalities $x < y$ or $x \leq y$. The order structure is closely related to the field structure. For example, when we use inequalities in elementary courses we frequently use the fact that if $x < y$ and $0 < z$, then $xz < yz$ (i.e., that inequalities can be multiplied through by positive numbers).

This structure, too, can be axiomatized and reduced to a small set of rules. Once again, these same rules can be found in other applications of mathematics. When these rules are added to the field axioms the result is called an *ordered field*.

The real number system is an ordered field, satisfying the four additional axioms. Here $a < b$ is now a statement that is either true or false. (Before $a + b$ and $a \cdot b$ were not statements, but elements of $\mathbb{R}$.)

O1 For any $a, b \in \mathbb{R}$ exactly one of the statements $a = b$, $a < b$ or $b < a$ is true.

O2 For any $a, b, c \in \mathbb{R}$ if $a < b$ is true and $b < c$ is true, then $a < c$ is true.

O3 For any $a, b \in \mathbb{R}$ if $a < b$ is true, then $a + c < b + c$ is also true for any $c \in \mathbb{R}$.

O4 For any $a, b \in \mathbb{R}$ if $a < b$ is true, then $a \cdot c < b \cdot c$ is also true for any $c \in \mathbb{R}$ for which $c > 0$.

Exercises

1.4.1 Using just the axioms, prove that $ad + bc < ac + bd$ if $a < b$ and $c < d$.

1.4.2 Show for every $n \in \mathbb{N}$ that $n^2 \geq n$.

1.4.3 Using just the axioms, prove the *arithmetic-geometric mean inequality*:

$$\sqrt{ab} \leq \frac{a+b}{2}$$

for any $a, b \in \mathbb{R}$ with $a > 0$ and $b > 0$. (Assume, for the moment, the existence of square roots.)

SEE NOTE 3

1.5 Bounds

Let E be some set of real numbers. There may or may not be a number M that is bigger than every number in the set E. If there is, we say that M is an upper bound for the set. If there is no upper bound, then the set is said to be *unbounded above* or to have no upper bound. This is a simple enough idea, but it is critical to an understanding of the real numbers and so we shall look more closely at it and give some precise definitions.

Definition 1.1: (Upper Bounds) Let E be a set of real numbers. A number M is said to be an *upper bound* for E if $x \leq M$ for all $x \in E$.

Definition 1.2: (Lower Bounds) Let E be a set of real numbers. A number m is said to be a *lower bound* for E if $m \leq x$ for all $x \in E$.

It is often important to note whether a set has bounds or not. A set that has an upper bound and a lower bound is called *bounded*.

A set can have many upper bounds. Indeed every number is an upper bound for the empty set $\emptyset$. A set may have no upper bounds. We can use the phrase "E is unbounded above" if there are no upper bounds. For some sets the most natural upper bound (from among the infinitely many to choose) is just the largest member of the set. This is called the maximum. Similarly, the most natural lower bound for some sets is the smallest member of the set, the minimum.

Definition 1.3: (Maximum) Let E be a set of real numbers. If there is a number M that belongs to E and is larger than every other member of E, then M is called the maximum of the set E and we write $M = \max E$.

Definition 1.4: (Minimum) Let E be a set of real numbers. If there is a number m that belongs to E and is smaller than every other member of E, then m is called the minimum of the set E and we write $m = \min E$.

Example 1.5: The interval

$$[0,1] = \{x : 0 \leq x \leq 1\}$$

has a maximum and a minimum. The maximum is 1 and 1 is also an upper bound for the set. (If a set has a maximum, then that number must certainly be an upper bound for the set.) Any number larger than 1 is also an upper bound. The number 0 is the minimum and also a lower bound. ◀

Example 1.6: The interval

$$(0,1) = \{x : 0 < x < 1\}$$

has no maximum and no minimum. At first glance some novices insist that the maximum should be 1 and the minimum 0 as before. But look at the definition. The maximum must be both an upper bound and also a member of the set. Here 1 and 0 are upper and lower bounds, respectively, but do not belong to the set. ◀

Example 1.7: The set $\mathbb{N}$ of natural numbers has a minimum but no maximum and no upper bounds at all. We would say that it is bounded below but not bounded above. ◀

1.6 Sups and Infs

Let us return to the subject of maxima and minima again. If E has a maximum, say M, then that maximum could be described by the statement

M is the least of all the upper bounds of E,

that is to say, M is the minimum of all the upper bounds. The most frequent language used here is "M is the least upper bound." It is possible for a set to have no maximum and yet be bounded above. In any example that comes to mind you will see that the set appears to have a least upper bound.

Example 1.8: The open interval $(0,1)$ has no maximum, but many upper bounds. Certainly 2 is an upper bound and so is 1. The least of all the upper bounds is the number 1. Note that 1 cannot be described as a maximum because it fails to be in the set. ◀

Definition 1.9: (Least Upper Bound/Supremum) Let E be a set of real numbers that is bounded above and nonempty. If M is the least of all the upper bounds, then M is said to be *the least upper bound of* E or *the supremum* of E and we write $M = \sup E$.

Definition 1.10: (Greatest Lower Bound/Infimum) Let E be a set of real numbers that is bounded below and nonempty. If m is the greatest of all the lower bounds of E, then m is said to be *the greatest lower bound of* E or *the infimum* of E and we write $M = \inf E$.

To complete the definition of $\inf E$ and $\sup E$ it is most convenient to be able write this expression even for $E = \emptyset$ or for unbounded sets. Thus we write

1. $\inf \emptyset = \infty$ and $\sup \emptyset = -\infty$.
2. If E is unbounded above, then $\sup E = \infty$.
3. If E is unbounded below, then $\inf E = -\infty$.

The Axiom of Completeness Any example of a nonempty set that you are able to visualize that has an upper bound will also have a least upper bound. Pages of examples might convince you that all nonempty sets bounded above must have a least upper bound. Indeed your intuition will forbid you to accept the idea that this could not always be the case. To prove such an assertion is not possible using only the axioms for an ordered field. Thus we shall assume one further axiom, known as the axiom of completeness.

> **Completeness Axiom** A nonempty set of real numbers that is bounded above has a least upper bound (i.e., if E is nonempty and bounded above, then $\sup E$ exists and is a real number).

This now is the totality of all the axioms we need to assume. We have assumed that $\mathbb{R}$ is a field with two operations of addition and multiplication, that $\mathbb{R}$ is an ordered field with an inequality relation "$<$", and finally that $\mathbb{R}$ is a complete ordered field. This is enough to characterize the real numbers and the phrase "complete ordered field" refers to the system of real numbers and to no other system. (We shall not prove this statement; see Exercise 1.11.3 for a discussion.)

Exercises

1.6.1 Show that a set of real numbers E is bounded if and only if there is a positive number r so that $|x| < r$ for all $x \in E$.

1.6.2 Find $\sup E$ and $\inf E$ and (where possible) $\max E$ and $\min E$ for the following examples of sets:

(a) $E = \mathbb{N}$
(b) $E = \mathbb{Z}$
(c) $E = \mathbb{Q}$
(d) $E = \mathbb{R}$
(e) $E = \{-3, 2, 5, 7\}$
(f) $E = \{x : x^2 < 2\}$
(g) $E = \{x : x^2 - x - 1 < 0\}$
(h) $E = \{1/n : n \in \mathbb{N}\}$
(i) $E = \{\sqrt[n]{n} : n \in \mathbb{N}\}$

1.6.3 Under what conditions does $\sup E = \max E$?

1.6.4 Show for every nonempty, finite set E that $\sup E = \max E$.
SEE NOTE 4

1.6.5 For every $x \in \mathbb{R}$ define

$$[x] = \max\{n \in \mathbb{Z} : n \leq x\}$$

called the *greatest integer function.* Show that this is well defined and sketch the graph of the function.

1.6.6 Let A be a set of real numbers and let $B = \{-x : x \in A\}$. Find a relation between $\max A$ and $\min B$ and between $\min A$ and $\max B$.

1.6.7 Let A be a set of real numbers and let $B = \{-x : x \in A\}$. Find a relation between $\sup A$ and $\inf B$ and between $\inf A$ and $\sup B$.

1.6.8 Let A be a set of real numbers and let $B = \{x + r : x \in A\}$ for some number r. Find a relation between $\sup A$ and $\sup B$.

1.6.9 Let A be a set of real numbers and let $B = \{xr : x \in A\}$ for some positive number r. Find a relation between $\sup A$ and $\sup B$. (What happens if r is negative?)

1.6.10 Let A and B be sets of real numbers such that $A \subset B$. Find a relation among $\inf A$, $\inf B$, $\sup A$, and $\sup B$.

1.6.11 Let A and B be sets of real numbers and write $C = A \cup B$. Find a relation among $\sup A$, $\sup B$, and $\sup C$.

1.6.12 Let A and B be sets of real numbers and write $C = A \cap B$. Find a relation among $\sup A$, $\sup B$, and $\sup C$.

1.6.13 Let A and B be sets of real numbers and write

$$C = \{x + y : x \in A,\ y \in B\}.$$

Find a relation among $\sup A$, $\sup B$, and $\sup C$.

1.6.14 Let A and B be sets of real numbers and write

$$C = \{x + y : x \in A,\ y \in B\}.$$

Find a relation among $\inf A$, $\inf B$, and $\inf C$.

1.6.15 Let A be a set of real numbers and write $A^2 = \{x^2 : x \in A\}$. Are there any relations you can find between the infs and sups of the two sets?

1.6.16 Let E be a set of real numbers. Show that x is not an upper bound of E if and only if there exists a number $e \in E$ such that $e > x$.

1.6.17 Let A be a set of real numbers. Show that a real number x is the supremum of A if and only if $a \leq x$ for all $a \in A$ and for every positive number ε there is an element $a' \in A$ such that $x - \varepsilon < a'$.

1.6.18 Formulate a condition analogous to the preceding exercise for an infimum.

1.6.19 Using the completeness axiom, show that every nonempty set E of real numbers that is bounded below has a greatest lower bound (i.e., $\inf E$ exists and is a real number).

1.6.20 A function is said to be *bounded* if its range is a bounded set. Give examples of functions $f : \mathbb{R} \to \mathbb{R}$ that are bounded and examples of such functions that are unbounded. Give an example of one that has the property that

$$\sup\{f(x) : x \in \mathbb{R}\}$$

is finite but $\max\{f(x) : x \in \mathbb{R}\}$ does not exist.

1.6.21 The rational numbers $\mathbb{Q}$ satisfy the axioms for an ordered field. Show that the completeness axiom would not be satisfied. That is show that this statement is false: Every nonempty set E of rational numbers that is bounded above has a least upper bound (i.e., $\sup E$ exists and is a rational number).

1.6.22 Let F be the set of all numbers of the form $x + \sqrt{2}y$, where x and y are rational numbers. Show that F has all the properties of an ordered field but does not have the completeness property.

1.6.23 Let A and B be nonempty sets of real numbers and let

$$\delta(A,B) = \inf\{|a-b| : a \in A,\ b \in B\}.$$

$\delta(A,B)$ is often called the "distance" between the sets A and B.

(a) Let $A = \mathbb{N}$ and $B = \mathbb{R} \setminus \mathbb{N}$. Compute $\delta(A,B)$

(b) If A and B are finite sets, what does $\delta(A,B)$ represent?

(c) Let $B = [0,1]$. What does the statement $\delta(\{x\},B) = 0$ mean for the point x?

(d) Let $B = (0,1)$. What does the statement $\delta(\{x\},B) = 0$ mean for the point x?

1.7 The Archimedean Property

There is an important relationship holding between the set of natural numbers $\mathbb{N}$ and the larger set of real numbers $\mathbb{R}$. Because we have a well-formed mental image of what the set of reals "looks like," this property is entirely intuitive and natural. It hardly seems that it would require a proof. It says that the set of natural numbers $\mathbb{N}$ has no upper bound (i.e., that there is no real number x so that $n \leq x$ for all $n = 1,2,3,\ldots$).

At first sight this seems to be a purely algebraic and order property of the reals. In fact it cannot be proved without invoking the completeness property of Section 1.6.

The property is named after the famous Greek mathematician known as Archimedes of Syracuse (287 B.C.–212 B.C.).[1]

Theorem 1.11 (Archimedean Property of $\mathbb{R}$) *The set of natural numbers $\mathbb{N}$ has no upper bound.*

Proof. The proof is obtained by contradiction. If the set $\mathbb{N}$ does have an upper bound, then it must have a least upper bound. Let $x = \sup \mathbb{N}$, supposing that such does exist as a finite real number. Then $n \leq x$ for all natural numbers n but $n \leq x-1$ cannot be true for all natural numbers n. Choose some natural number m with $m > x-1$. Then $m+1$ is also an natural number and $m+1 > x$. But that cannot be so since we defined x as the supremum. From this contradiction the theorem follows. ■

The archimedean theorem has some consequences that have a great impact on how we must think of the real numbers.

1. No matter how large a real number x is given, there is always a natural number n larger.

[1] Archimedes seems to be the archetypical absent-minded mathematician. The historian Plutarch tells of his death at the hand of an invading army: "As fate would have it, Archimedes was intent on working out some problem by a diagram, and having fixed both his mind and eyes upon the subject of his speculation, he did not notice the entry of the Romans nor that the city was taken. In this transport of study a soldier unexpectedly came up to him and commanded that he accompany him. When he declined to do this before he had finished his problem, the enraged soldier drew his sword and ran him through." For this biographical detail and many others on all the mathematicians in this book consult *http://www-history.mcs.st-and.ac.uk/history*.

2. Given any positive number y, no matter how large, and any positive number x, no matter how small, one can add x to itself sufficiently many times so that the result exceeds y (i.e., $nx > y$ for some $n \in \mathbb{N}$).

3. Given any positive number x, no matter how small, one can always find a fraction $1/n$ with n a natural number that is smaller (i.e., so that $1/n < x$).

Each of these is a consequence of the archimedean theorem, and the archimedean theorem in turn can be derived from any one of these.

Exercises

1.7.1 Using the archimedean theorem, prove each of the three statements that follow the proof of the archimedean theorem.

1.7.2 Suppose that it is true that for each $x > 0$ there is an $n \in \mathbb{N}$ so that $1/n < x$. Prove the archimedean theorem using this assumption.

1.7.3 Without using the archimedean theorem, show that for each $x > 0$ there is an $n \in \mathbb{N}$ so that $1/n < x$.
See Note 5

1.7.4 Let x be any real number. Show that there is a $m \in \mathbb{Z}$ so that

$$m \leq x < m+1.$$

Show that m is unique.

1.7.5 The mathematician Leibniz based his calculus on the assumption that there were "infinitesimals," positive real numbers that are extremely small—smaller than all positive rational numbers certainly. Some calculus students also believe, apparently, in the existence of such numbers since they can imagine a number that is "just next to zero." Is there a positive real number smaller than all positive rational numbers?

1.7.6 The archimedean property asserts that if $x > 0$, then there is a natural number N so that $1/N < x$. The proof requires the completeness axiom. Give a proof that does not use the completeness axiom that works for x rational. Find a proof that is valid for $x = \sqrt{y}$, where y is rational.

1.7.7 In Section 1.2 we made much of the fact that there is a number whose square is 2 and so $\sqrt{2}$ does exist as a real number. Show that

$$\alpha = \sup\{x \in \mathbb{R} : x^2 < 2\}$$

exists as a real number and that $\alpha^2 = 2$.
See Note 6

1.8 Inductive Property of $\mathbb{N}$

Since the natural numbers are included in the set of real numbers there are further important properties of $\mathbb{N}$ that can be deduced from the axioms. The most important of these is the principle of induction. This is the basis for the technique of proof known as induction, which is often used in this text. For an elementary account and some practice, see Section A.8 in the appendix.

We first prove a statement that is equivalent.

Theorem 1.12 (Well-Ordering Property) *Every nonempty subset of $\mathbb{N}$ has a smallest element.*

Proof. Let $S \subset \mathbb{N}$ and $S \neq \emptyset$. Then $\alpha = \inf S$ must exist and be a real number since S is bounded below. If $\alpha \in S$, then we are done since we have found a minimal element.

Suppose not. Then, while α is the greatest lower bound of S, α is not a minimum. There must be an element of S that is smaller than $\alpha + 1$ since α is the greatest lower bound of S. That element cannot be α since we have assumed that $\alpha \notin S$. Thus we have found $x \in S$ with

$$\alpha < x < \alpha + 1.$$

Now x is not a lower bound of S, since it is greater than the greatest lower bound of S, so there must be yet another element y of S such that

$$\alpha < y < x < \alpha + 1.$$

But now we have reached an impossibility, for x and y are in S and both natural numbers, but $0 < x - y < 1$, which cannot happen. From this contradiction the proof now follows. ■

Now we can state and prove the principle of induction.

Theorem 1.13 (Principle of Induction) *Let $S \subset \mathbb{N}$ so that $1 \in S$ and, for every natural number n, if $n \in S$ then so also is $n+1$. Then $S = \mathbb{N}$.*

Proof. Let $E = \mathbb{N} \setminus S$. We claim that $E = \emptyset$ and then it follows that $S = \mathbb{N}$ proving the theorem. Suppose not (i.e., suppose $E \neq \emptyset$). By Theorem 1.12 there is a first element α of E. Can $\alpha = 1$? No, because $1 \in S$ by hypothesis. Thus $\alpha - 1$ is also a natural number and, since it cannot be in E it must be in S. By hypothesis it follows that $\alpha = (\alpha - 1) + 1$ must be in S. But it is in E. This is impossible and so we have obtained a contradiction, proving our theorem. ■

Exercises

1.8.1 Show that any bounded, nonempty set of natural numbers has a maximal element.

1.8.2 Show that any bounded, nonempty subset of $\mathbb{Z}$ has a maximum and a minimum.

1.8.3 For further exercises on proving statements using induction as a method, see Section A.8.

1.9 The Rational Numbers Are Dense

There is an important relationship holding between the set of rational numbers $\mathbb{Q}$ and the larger set of real numbers $\mathbb{R}$. The rational numbers are dense. They make an appearance in every interval; there are no gaps, no intervals that miss having rational numbers.

For practical purposes this has great consequences. We need never actually compute with arbitrary real numbers, since close by are rational numbers that can be used. Thus,

while π is irrational, in routine computations with a practical view any nearby fraction might do. At various times people have used 3, $22/7$, and 3.14159, for example.

For theoretical reasons this fact is of great importance too. It allows many arguments to replace a consideration of the set of real numbers with the smaller set of rationals. Since every real is as close as we please to a rational and since the rationals can be carefully described and easily worked with, many simplifications are allowed.

Definition 1.14: (Dense Sets) A set E of real numbers is said to be *dense* (or *dense in* $\mathbb{R}$) if every interval (a,b) contains a point of E.

Theorem 1.15: *The set $\mathbb{Q}$ of rational numbers is dense.*

Proof. Let $x < y$ and consider the interval (x,y). We must find a rational number inside this interval.

By the archimedean theorem, Theorem 1.11, there is a natural number

$$n > \frac{1}{y-x}.$$

This means that $ny > nx+1$.

Let m be chosen as the integer just less than $nx+1$; more precisely (using Exercise 1.7.4), find $m \in \mathbb{Z}$ so that

$$m \leq nx+1 < m+1.$$

Now some arithmetic on these inequalities shows that

$$m-1 \leq nx < ny$$

and then

$$x < \frac{m}{n} \leq x + \frac{1}{n} < y$$

thus exhibiting a rational number m/n in the interval (x,y). ■

Exercises

1.9.1 Show that the definition of "dense" could be given as

> A set E of real numbers is said to be *dense* if every interval (a,b) contains infinitely many points of E.

1.9.2 Find a rational number between $\sqrt{10}$ and π.

1.9.3 If a set E is dense, what can you conclude about a set $A \supset E$?

1.9.4 If a set E is dense, what can you conclude about the set $\mathbb{R} \setminus E$?

1.9.5 If two sets E_1 and E_2 are dense, what can you conclude about the set $E_1 \cap E_2$?

1.9.6 Show that the *dyadic rationals* (i.e., rational numbers of the form $m/2^n$ for $m \in \mathbb{Z}$, $n \in \mathbb{N}$) are dense.

1.9.7 Are the numbers of the form

$$\pm m/2^{100}$$

for $m \in \mathbb{N}$ dense? What is the length of the largest interval that contains no such number?

1.9.8 Show that the numbers of the form

$$\pm m\sqrt{2}/n$$

for $m, n \in \mathbb{N}$ are dense.

SEE NOTE 7

1.10 The Metric Structure of $\mathbb{R}$

In addition to the algebraic and order structure of the real numbers, we need to make measurements. We need to describe distances between points. These are the metric properties of the reals, to borrow a term from the Greek for measure (*metron*).

As usual, the distance between a point x and another point y is either $x-y$ or $y-x$ depending on which is positive. Thus the distance between 3 and -4 is 7. The distance between π and $\sqrt{10}$ is $\sqrt{10}-\pi$. To describe this in general requires the absolute value function which simply makes a choice between positive and negative.

Definition 1.16: (Absolute Value) For any real number x write

$$|x| = x \text{ if } x \geq 0$$

and

$$|x| = -x \text{ if } x < 0\,.$$

(Beginners tend to think of the absolute value function as "stripping off the negative sign," but the example

$$|\pi - \sqrt{10}| = \sqrt{10} - \pi$$

shows that this is a limited viewpoint.)

Properties of the Absolute Value Since the absolute value is defined directly in terms of inequalities (i.e., the choice $x \geq 0$ or $x < 0$), there are a number of properties that can be proved directly from properties of inequalities. These properties are used routinely and the student will need to have a complete mastery of them.

Theorem 1.17: *The absolute value function has the following properties:*

1. *For any $x \in \mathbb{R}$, $-|x| \leq x \leq |x|$.*
2. *For any $x, y \in \mathbb{R}$, $|xy| = |x|\,|y|$.*
3. *For any $x, y \in \mathbb{R}$, $|x+y| \leq |x| + |y|$.*
4. *For any $x, y \in \mathbb{R}$, $|x| - |y| \leq |x-y|$ and $|y| - |x| \leq |x-y|$.*

Distances on the Real Line Using the absolute value function we can define the distance function or metric.

Definition 1.18: (Distance) The distance between two real numbers x and y is

$$d(x,y) = |x-y|.$$

We hardly ever use the notation $d(x,y)$ in elementary analysis, preferring to write $|x-y|$ even while we are thinking of this as the distance between the two points. Thus if a sequence of points $x_1, x_2, x_3, \ldots$ is growing ever closer to a point c, we should perhaps describe $d(x_n,c)$ as getting smaller and smaller, thus emphasizing that the distances are shrinking; more often we would simply write $|x_n - c|$ and expect you to interpret this as a distance.

Properties of the Distance Function The main properties of the distance function are just interpretations of the absolute value function. Expressed in the language of a distance function, they are geometrically very intuitive:

1. $d(x,y) \geq 0$

   (all distances are positive or zero).

2. $d(x,y) = 0$ if and only if $x = y$

   (different points are at positive distance apart).

3. $d(x,y) = d(y,x)$

   (distance is symmetric, that is the distance from x to y is the same as from y to x)).

4. $d(x,y) \leq d(x,z) + d(z,y)$

   (the triangle inequality, that is it is no longer to go directly from x to y than to go from x to z and then to y).

In Chapter 13 we will study general structures called metric spaces, where exactly such a notion of distance satisfying these four properties is used. For now we prefer to rewrite these properties in the language of the absolute value, where they lose some of their intuitive appeal. But it is in this form that we are likely to use them.

1. $|a| \geq 0$.

2. $|a| = 0$ if and only if $a = 0$.

3. $|a| = |-a|$.

4. $|a+b| \leq |a| + |b|$ (the triangle inequality).

Exercises

1.10.1 Show that $|x| = \max\{x, -x\}$.

1.10.2 Show that $\max\{x,y\} = |x-y|/2 + (x+y)/2$. What expression would give $\min\{x,y\}$?

1.10.3 Show that the inequalities $|x-a| < \varepsilon$ and

$$a - \varepsilon < x < a + \varepsilon$$

are equivalent.

1.10.4 Show that if $\alpha < x < \beta$ and $\alpha < y < \beta$, then $|x-y| < \beta - \alpha$ and interpret this geometrically as a statement about the interval (α, β).

1.10.5 Show that $||x| - |y|| \leq |x-y|$ assuming the triangle inequality (i.e., that $|a+b| \leq |a| + |b|$). This inequality is also called the triangle inequality.

1.10.6 Under what conditions is it true that $|x+y| = |x| + |y|$?

1.10.7 Under what conditions is it true that

$$|x-y| + |y-z| = |x-z|?$$

1.10.8 Show that

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$$

for any numbers $x_1, x_2, \ldots, x_n$.

1.10.9 Let E be a set of real numbers and let $A = \{|x| : x \in E\}$. What relations can you find between the infs and sups of the two sets?

1.10.10 Find the inf and sup of the set $\{x : |2x + \pi| < \sqrt{2}\}$.

1.11 Challenging Problems for Chapter 1

1.11.1 The complex numbers $\mathbb{C}$ are defined as equal to the set of all ordered pairs of real numbers subject to these operations:

$$(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2)$$

and

$$(a_1, b_1) \cdot (a_2, b_2) = (a_1 a_2 - b_1 b_2, a_1 b_2 + a_2 b_1).$$

(a) Show that $\mathbb{C}$ is a field.

(b) What are the additive and multiplicative identity elements?

(c) What are the additive and multiplicative inverses of an element (a, b)?

(d) Solve $(a, b)^2 = (1, 0)$ in $\mathbb{C}$.

(e) We identify $\mathbb{R}$ with a subset of $\mathbb{C}$ by identifying the elements $x \in \mathbb{R}$ with the element $(x, 0)$ in $\mathbb{C}$. Explain how this can be interpreted as saying that "$\mathbb{R}$ is a subfield of $\mathbb{C}$."

(f) Show that there is an element $i \in \mathbb{C}$ with $i^2 = -1$ so that every element $z \in \mathbb{C}$ can be written as $z = x + iy$ for $x, y \in \mathbb{R}$.

(g) Explain why the equation $x^2 + x + 1 = 0$ has no solution in $\mathbb{R}$ but two solutions in $\mathbb{C}$.

1.11.2 Can an order be defined on the field $\mathbb{C}$ of Exercise 1.11.1 in such a way so to make it an ordered field?

1.11.3 The statement that every complete ordered field "is" the real number system means the following. Suppose that F is a nonempty set with operations of addition "$+$" and multiplication "$\cdot$" and an order relation "$<$" that satisfies all the axioms of an ordered field and also the axiom of completeness. Then there is a one-to-one onto function $f : \mathbb{R} \to F$ that has the following properties:

(a) $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

(b) $f(x \cdot y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$.

(c) $f(x) < f(y)$ if and only if $x < y$ for $x, y \in \mathbb{R}$.

Thus, in a certain sense, F and $\mathbb{R}$ are essentially the same object. Attempt a proof of this statement. [Note that $x+y$ for $x, y \in \mathbb{R}$ refers to the addition in the reals whereas $f(x)+f(y)$ refers to the addition in the set F.]

1.11.4 We have assumed in the text that the set $\mathbb{N}$ is obviously contained in $\mathbb{R}$. After all, 1 is a real number (it's in the axioms), 2 is just $1+1$ and so real, 3 is $2+1$ etc. In that way we have been able to prove the material of Section 1.8. But there is a logical flaw here. We would need induction really to define $\mathbb{N}$ in this way (and not just say "etc."). Here is a set of exercises that would remedy that for students with some background in set manipulations.

(a) Define a set $S \subset \mathbb{R}$ to be *inductive* if $1 \in S$ and if $x \in S$ implies that $x+1 \in S$. Show that $\mathbb{R}$ is inductive.

(b) Show that there is a smallest inductive set by showing that the intersection of the family of all inductive sets is itself inductive.

(c) Define $\mathbb{N}$ to be that smallest inductive set.

(d) Prove Theorem 1.13 now. (That is, show that any set S with the property stated there is inductive and conclude that $S = \mathbb{N}$.)

(e) Prove Theorem 1.12 now. (That is, with this definition of $\mathbb{N}$ prove the well-ordering property.)

1.11.5 Use this definition of "dense in a set" to answer the following questions:

> A set E of real numbers is said to be *dense in a set* A if every interval (a,b) that contains a point of A also contains a point of E.

(a) Show that dense in the set of all reals is the same as dense.

(b) Give an example of a set E dense in $\mathbb{N}$ but with $E \cap \mathbb{N} = \emptyset$.

(c) Show that the irrationals are dense in the rationals. (A real number is *irrational* if it is not rational, that is if it belongs to $\mathbb{R}$ but not to $\mathbb{Q}$.)

(d) Show that the rationals are dense in the irrationals.

(e) What property does a set E have that is equivalent to the assertion that $\mathbb{R} \setminus E$ is dense in E?

1.11.6 Let G be a subgroup of the real numbers under addition (i.e., if x and y are in G, then $x+y \in G$ and $-x \in G$). Show that either G is a dense subset of $\mathbb{R}$ or else there is a real number α so that

$$G = \{n\alpha : n = 0, \pm 1, \pm 2, \pm 3, \dots\}.$$

SEE NOTE 8

Notes

[1]Exercise 1.3.3. Let F be the set of all numbers of the form $x+y\sqrt{2}$ where $x, y \in \mathbb{Q}$. Again to be sure that nine properties of a field hold it is enough to check, here, that $a+b$ and $a \cdot b$ are in F if both a and b are.

[2]Exercise 1.3.5. As a first step define what x^2 and $2x$ really mean. In fact, define 2. (It would be defined as $2 = 1+1$ since 1 and addition are defined in the field axioms.) Then multiply $(x+1)\cdot(x+1)$ using only the rules given here. Since your proof uses only the field axioms, it must be valid in any situation in which these axioms are true, not just for $\mathbb{R}$.

[3]Exercise 1.4.3. Suppose $a > 0$ and $b > 0$ and $a \neq b$. Establish that $\sqrt{a} \neq \sqrt{b}$. Establish that

$$(\sqrt{a}-\sqrt{b})^2 > 0.$$

Carry on. What have you proved? Now what if $a = b$?

[4]Exercise 1.6.4. You can use induction on the size of E, that is, prove for every natural number n that if E has n elements, then

$$\sup E = \max E.$$

[5]Exercise 1.7.3. Suppose not, then the set

$$\{1/n : n = 1,2,3,\dots\}$$

has a positive lower bound, etc. You will have to use the existence of a greatest lower bound.

[6]Exercise 1.7.7. Not that easy to show. Rule out the possibilities $\alpha^2 < 2$ and $\alpha^2 > 2$ using the archimedean property to assist.

[7]Exercise 1.9.8. To find a number in (x,y), find a rational in $(x//\sqrt{2}, y//\sqrt{2})$. Conclude from this that the set of all (irrational) numbers of the form $\pm m\sqrt{2}/n$ is dense.

[8]Exercise 1.11.6. If $G = \{0\}$, then take $\alpha = 0$. If not, let $\alpha = \inf G \cap (0,\infty)$. Case 1: If $\alpha = 0$ show that G is dense. Case 2: If $\alpha > 0$ show that

$$G = \{n\alpha : n = 0, \pm 1, \pm 2, \pm 3, \dots\}.$$

For case 1 consider an interval (r,s) with $r < s$. We wish to find a member of G in that interval. To keep the argument simple just consider, for the moment, the situation in which $0 < r < s$. Choose $g \in G$ with $0 < g < s-r$. The set

$$M = \{n \in \mathbb{N} : ng \geq s\}$$

is nonempty (why?) and so there is a minimal element m in M (why?). Now check that $(m-1)g$ is in G and inside the interval (r,s).

Chapter 2

SEQUENCES

2.1 Introduction

Let us start our discussion with a method for solving equations that originated with Newton in 1669. To solve an equation $f(x) = 0$ the method proposes the introduction of a new function

$$F(x) = x - \frac{f(x)}{f'(x)}.$$

We begin with a guess at a solution of $f(x) = 0$, say x_1 and compute $x_2 = F(x_1)$ in the hopes that x_2 is closer to a solution than x_1 was. The process is repeated so that $x_3 = F(x_2)$, $x_4 = F(x_3)$, $x_5 = F(x_4)$, ... and so on until the desired accuracy is reached. Processes of this type have been known for at least 3500 years although not in such a modern notation.

We illustrate by finding an approximate value for $\sqrt{2}$ this way. We solve the equation $f(x) = x^2 - 2 = 0$ by computing the function

$$F(x) = x - \frac{f(x)}{f'(x)} = x - \frac{x^2 - 2}{2x}$$

and using it to improve our guess. A first (very crude) guess of $x_1 = 1$ will produce the following list of values for our subsequent steps in the procedure. We have retained 60 digits in the decimal expansions to show how this is working:

$$x_1 = 1.000000000000000000000000000000000000000000000000000000000000$$

$$x_2 = 1.500000000000000000000000000000000000000000000000000000000000$$

$$x_3 = 1.416666666666666666666666666666666666666666666666666666666667$$

$$x_4 = 1.414215686274509803921568627450980392156862745098039215686280$$

$$x_5 = 1.414213562374689910626295578890134910116559622115744044584900$$

$$x_6 = 1.414213562373095048801689623502530243614981925776197428498290$$

$$x_7 = 1.414213562373095048801688724209698078569671875377234001561010.$$

To compare, here is the value of the true solution $\sqrt{2}$, computed in a different fashion to the same number of digits:

$$\sqrt{2} = 1.414213562373095048801688724209698078569671875376948073176680.$$

Note that after only four steps the procedure gives a value differing from the true value only in the sixth decimal place, and all subsequent values remain this close. A convenient way of expressing this is to write that

$$|x_n - \sqrt{2}| < 10^{-5} \text{ for all } n \geq 4.$$

By the seventh step, things are going even better and we can claim that

$$|x_n - \sqrt{2}| < 10^{-47} \text{ for all } n \geq 7.$$

It is inconceivable that anyone would require any further accuracy for any practical considerations. The error after the sixth step cannot exceed 10^{-47}, which is a tiny number. Even so, as mathematicians we can ask what may seem an entirely impractical sort of question. Can this accuracy of approximation continue forever? Is it possible that, if we wait long enough, we can find an approximation to $\sqrt{2}$ with *any* degree of accuracy?

Expressed more formally, if we are given a positive number ε (we call it epsilon to suggest that it measures an *error*) no matter how small, can we find a stage in this procedure so that the value computed and all subsequent values are closer to $\sqrt{2}$ than ε? In symbols, is there an integer n_0 (which will depend on just how small ε is) that is large enough so that

$$|x_n - \sqrt{2}| < \varepsilon \text{ for all } n \geq n_0?$$

If this is true then this sequence has a remarkable property. It is not merely in its first few terms a convenient way of computing $\sqrt{2}$ to some accuracy; the sequence truly represents the number $\sqrt{2}$ itself, and it cannot represent any other number. We shall say that the sequence converges to $\sqrt{2}$ and write

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

This is the beginning of the theory of convergence that is central to analysis. If mathematicians had never considered the ultimate behavior of such sequences and had contented themselves with using only the first few terms for practical computations, there would have been no subject known as analysis. These ideas lead, as you might imagine, to an ideal world of infinite precision, where sequences are not merely useful gadgets for getting good computations but are precise tools in discussing real numbers. From the theory of sequences and their convergence properties has developed a vast world of beautiful and useful mathematics.

For the student approaching this material for the first time this is a critical test. All of analysis, both pure and applied, rests on an understanding of limits. What you learn in this chapter will offer a foundation for all the rest that you will have to learn later.

2.2 Sequences

A sequence (of real numbers, of sets, of functions, of anything) is simply a list. There is a first element in the list, a second element, a third element, and so on continuing in an order forever. In mathematics a finite list is not called a sequence; a sequence must continue without interruption.

For a more formal definition notice that the natural numbers are playing a key role here. Every item in the sequence (the list) can be labeled by its position; label the first item with a "1," the second with a "2," and so on. Seen this way a sequence is merely then a function mapping the natural numbers $\mathbb{N}$ into some set. We state this as a definition. Since this chapter is exclusively about sequences of real numbers, the definition considers just this situation.

Definition 2.1: By a *sequence of real numbers* we mean a function

$$f : \mathbb{N} \to \mathbb{R}.$$

Thus the sequence *is* the function. Even so, we usually return to the list idea and write out the sequence f as

$$f(1), f(2), f(3), \ldots, f(n), \ldots$$

with the ellipsis (i.e., the three dots) indicating that the list is to continue in this fashion. The function values $f(1)$, $f(2)$, $f(3)$, ... are called the *terms* of the sequence. When it is not confusing we will refer to such a sequence using the expression

$$\{f(n)\}$$

(with the understanding that the index n ranges over all of the natural numbers).

If we need to return to the formality of functions we do, but try to keep the intuitive notion of a sequence as an unending list in mind. While computer scientists much prefer the function notation, mathematicians have become more accustomed to a subscript notation and would rather have the terms of the preceding sequence rendered as

$$f_1, f_2, f_3, \ldots, f_n, \ldots \quad \text{or} \quad \{f_n\}.$$

In this chapter we study sequences of real numbers. Later on we will encounter the same word applied to other lists of objects (e.g., sequences of intervals, sequences of sets, sequences of functions. In all cases the word sequence simply indicates a list of objects).

2.2.1 Sequence Examples

In order to specify some sequence we need to communicate what every term in the sequence is. For example, the sequence of even integers

$$2, 4, 6, 8, 10, \ldots$$

could be communicated in precisely that way: "Consider the sequence of even integers." Perhaps more direct would be to give a formula for all of the terms in the sequence: "Consider the sequence whose nth term is $x_n = 2n$." Or we could note that the sequence starts with 2 and then all the rest of the terms are obtained by adding 2 to the previous term: "Consider the sequence whose first term is 2 and whose nth term is 2 added to the $(n-1)$st term," that is,

$$x_n = 2 + x_{n-1}.$$

Often an explicit formula is best. Frequently though, a formula relating the nth term to some preceding term is preferable. Such formulas are called *recursion formulas* and would usually be more efficient if a computer is used to generate the terms.

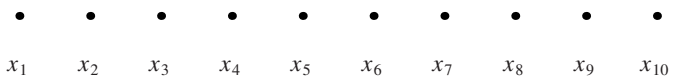


Figure 2.1. An arithmetic progression.

Arithmetic Progressions The simplest types of sequences are those in which each term is obtained from the preceding by adding a fixed amount. These are called *arithmetic progressions*. The sequence

$$c, c+d, c+2d, c+3d, c+4d, \ldots, c+(n-1)d, \ldots$$

is the most general arithmetic progression. The number d is called the *common difference*.

Every arithmetic progression could be given by a formula

$$x_n = c+(n-1)d$$

or a recursion formula

$$x_1 = c \quad x_n = x_{n-1}+d.$$

Note that the explicit formula is of the form $x_n = f(n)$, where f is a linear function, $f(x) = dx+b$ for some b. Figure 2.1 shows the points of an arithmetic progression plotted on the line. If, instead, you plot the points (n, x_n) you will find that they all lie on a straight line with slope d.

Geometric Progressions. A variant on the arithmetic progression is obtained by replacing the addition of a fixed amount by the multiplication by a fixed amount. These sequences are called *geometric progressions*. The sequence

$$c, cr, cr^2, cr^3, cr^4, \ldots, cr^{n-1}, \ldots$$

is the most general geometric progression. The number r is called the *common ratio*.

Every geometric progression could be given by a formula

$$x_n = cr^{n-1}$$

or a recursion formula

$$x_1 = c \quad x_n = rx_{n-1}.$$

Note that the explicit formula is of the form $x_n = f(n)$, where f is an exponential function $f(x) = br^x$ for some b. Figure 2.2 shows the points of a geometric progression plotted on the line. Alternatively, plot the points (n, x_n) and you will find that they all lie on the graph of an exponential function. If $c > 0$ and the common ratio r is larger than 1, the terms increase in size, becoming extremely large. If $0 < r < 1$, the terms decrease in size, getting smaller and smaller. (See Figure 2.2.)

Iteration The examples of an arithmetic progression and a geometric progression are special cases of a process called *iteration*. So too is the sequence generated by Newton's method in the introduction to this chapter.

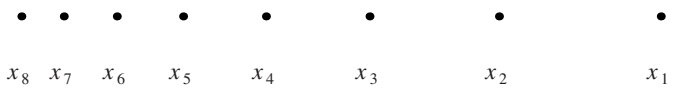


Figure 2.2. A geometric progression.

Let f be some function. Start the sequence $x_1, x_2, x_3, \ldots$ by assigning some value in the domain of f, say $x_1 = c$. All subsequent values are now obtained by feeding these values through the function repeatedly:

$$c, f(c), f(f(c)), f(f(f(c))), f(f(f(f(c)))), \ldots.$$

As long as all these values remain in the domain of the function f, the process can continue indefinitely and defines a sequence. If f is a function of the form $f(x) = x + b$, then the result is an arithmetic progression. If f is a function of the form $f(x) = ax$, then the result is a geometric progression.

A recursion formula best expresses this process and would offer the best way of writing a computer program to compute the sequence:

$$x_1 = c \quad x_n = f(x_{n-1}).$$

Sequence of Partial Sums. If a sequence

$$x_1, x_2, x_3, x_4, \ldots$$

is given, we can construct a new sequence by adding the terms of the old one:

$$s_1 = x_1$$

$$s_2 = x_1 + x_2$$

$$s_3 = x_1 + x_2 + x_3$$

$$s_4 = x_1 + x_2 + x_3 + x_4$$

and continuing in this way. The process can also be described by a recursion formula:

$$s_1 = x_1\,, \quad s_n = s_{n-1} + x_n.$$

The new sequence is called the *sequence of partial sums* of the old sequence $\{x_n\}$. We shall study such sequences in considerable depth in the next chapter.

For a particular example we could use $x_n = 1/n$ and the sequence of partial sums could be written as

$$s_n = 1 + 1/2 + 1/3 + \cdots + 1/n.$$

Is there a more attractive and simpler formula for s_n? The answer is no.

Example 2.2: The examples, given so far, are of a general nature and describe many sequences that we will encounter in analysis. But a sequence is just a list of numbers and need not be defined in any manner quite so systematic. For example, consider the sequence defined by $a_n = 1$ if n is divisible by three, $a_n = n$ if n is one more than a

multiple of three, and $a_n = -2^n$ if n is two more than a multiple of three. The first few terms are evidently

$$1, -4, 1, 4, -32, 1, \ldots.$$

What would be the next three terms? ◀

Exercises

2.2.1 Let a sequence be defined by the phrase "consider the sequence of prime numbers $2,3,5,7,11,13\ldots$". Are you sure that this defines a sequence?

2.2.2 On IQ tests one frequently encounters statements such as "what is the next term in the sequence 3, 1, 4, 1, 5, ...?". In terms of our definition of a sequence is this correct usage? (By the way, what do you suppose the next term in the sequence might be?)
SEE NOTE 9

2.2.3 Give two different formulas (for two different sequences) that generate a sequence whose first four terms are 2, 4, 6, 8.
SEE NOTE 10

2.2.4 Give a formula that generates a sequence whose first five terms are 2, 4, 6, 8, π.

2.2.5 The examples listed here are the first few terms of a sequence that is either an arithmetic progression or a geometric progression. What is the next term in the sequence? Give a general formula for the sequence.

(a) 7, 4, 1, ...

(b) .1, .01, .001, ...

(c) 2, $\sqrt{2}$, 1, ...

2.2.6 Consider the sequence defined recursively by

$$x_1 = \sqrt{2}, \quad x_n = \sqrt{2} + x_{n-1}.$$

Find an explicit formula for the nth term.

2.2.7 Consider the sequence defined recursively by

$$x_1 = \sqrt{2}, \quad x_n = \sqrt{2}x_{n-1}.$$

Find an explicit formula for the nth term.

2.2.8 Consider the sequence defined recursively by

$$x_1 = \sqrt{2}, \quad x_n = \sqrt{2 + x_{n-1}}.$$

Show, by induction, that $x_n < 2$ for all n.

2.2.9 Consider the sequence defined recursively by

$$x_1 = \sqrt{2}, \quad x_n = \sqrt{2 + x_{n-1}}.$$

Show, by induction, that $x_n < x_{n+1}$ for all n.

2.2.10 The sequence defined recursively by

$$f_1 = 1, \; f_2 = 1, \quad f_{n+2} = f_n + f_{n+1}$$

is called the *Fibonacci sequence*. It is possible to find an explicit formula for this sequence. Give it a try.
SEE NOTE 11

2.3 Countable Sets

✂ *Enrichment section. May be omitted.*

A sequence of real numbers, formally, is a function whose domain is the set $\mathbb{N}$ of natural numbers and whose range is a subset of the reals $\mathbb{R}$. What sets might be the range of some sequence? To put it another way, what sets can have their elements arranged into an unending list? Are there sets that cannot be arranged into a list?

The arrangement of a collection of objects into a list is sometimes called an *enumeration*. Thus another way of phrasing this question is to ask what sets of real numbers can be *enumerated*?

The set of natural numbers is already arranged into a list in its natural order. The set of integers (including 0 and the negative integers) is not usually presented in the form of a list but can easily be so presented, as the following scheme suggests:

$$0,1,-1,2,-2,3,-3,4,-4,5,-5,6,-6,7,-7,\ldots.$$

Example 2.3: The rational numbers can also be listed but this is quite remarkable, for at first sight no reasonable way of ordering them into a sequence seems likely to be possible. The usual order of the rationals in the reals is of little help.

To find such a scheme define the "rank" of a rational number m/n in its lowest terms (with $n \geq 1$) to be $|m|+n$. Now begin making a finite list of all the rational numbers at each rank; list these from smallest to largest. For example, at rank 1 we would have only the rational number $0/1$. At rank 2 we would have only the rational numbers $-1/1$, $1/1$. At rank 3 we would have only the rational numbers $-2/1$, $-1/2$, $1/2$, $2/1$. Carry on in this fashion through all the ranks. Now construct the final list by concatenating these shorter lists in order of the ranks:

$$0/1,-1/1,1/1,-2/1,-1/2,1/2,2/1,\ldots.$$

The range of this sequence is the set of all rational numbers. ◀

Your first impression might be that few sets would be able to be the range of a sequence. But having seen in Example 2.3 that even the set of rational numbers $\mathbb{Q}$ that is seemingly so large can be listed, it might then appear that all sets can be so listed. After all, can you conceive of a set that is "larger" than the rationals in some way that would stop it being listed? The remarkable fact that there are sets that cannot be arranged to form the elements of some sequence was proved by Georg Cantor (1845–1918). This proof is essentially his original proof. (Note that this requires some familiarity with infinite decimal expansions; the exercises review what is needed.)

Theorem 2.4 (Cantor) *No interval (a,b) of real numbers can be the range of some sequence.*

Proof. It is enough to prove this for the interval $(0,1)$ since there is nothing special about it (see Exercise 2.3.1). The proof is a proof by contradiction. We suppose that the theorem is false and that there is a sequence $\{s_n\}$ so that every number in the interval $(0,1)$ appears at least once in the sequence. We obtain a contradiction by showing that

this cannot be so. We shall use the sequence $\{s_n\}$ to find a number c in the interval $(0,1)$ so that $s_n \neq c$ for all n.

Each of the points $s_1, s_2, s_3 \ldots$ in our sequence is a number between 0 and 1 and so can be written as a decimal fraction. If we write this sequence out in decimal notation it might look like

$$s_1 = 0.x_{11}x_{12}x_{13}x_{14}x_{15}x_{16}\ldots$$
$$s_2 = 0.x_{21}x_{22}x_{23}x_{24}x_{25}x_{26}\ldots$$
$$s_3 = 0.x_{31}x_{32}x_{33}x_{34}x_{35}x_{36}\ldots$$

etc. Now it is easy to find a number that is not in the list. Construct

$$c = 0.c_1c_2c_3c_4c_5c_6\ldots$$

by choosing c_i to be either 5 or 6 whichever is different from x_{ii}. This number cannot be equal to any of the listed numbers $s_1, s_2, s_3 \ldots$ since c and s_i differ in the ith position of their decimal expansions. This gives us our contradiction and so proves the theorem. ■

Definition 2.5: (Countable) A nonempty set S of real numbers is said to be *countable* if there is a sequence of real numbers whose range is the set S.

In the language of this definition then we can see that (1) any finite set is countable, (2) the natural numbers and the integers are countable, (3) the rational numbers are countable, and (4) no interval of real numbers is countable. By convention we also say that the empty set $\emptyset$ is countable.

Exercises

2.3.1 Show that, once it is known that the interval $(0,1)$ cannot be expressed as the range of some sequence, it follows that any interval (a,b), $[a,b)$, $(a,b]$, or $[a,b]$ has the same property.
SEE NOTE 12

2.3.2 Some novices, on reading the proof of Cantor's theorem, say "Why can't you just put the number c that you found at the front of the list." What is your rejoinder?

2.3.3 A set (any set of objects) is said to be *countable* if it is either finite or there is an enumeration (list) of the set. Show that the following properties hold for arbitrary countable sets:

(a) All subsets of countable sets are countable.

(b) Any union of a pair of countable sets is countable.

(c) All finite sets are countable.

2.3.4 Show that the following property holds for countable sets: If

$$S_1, S_2, S_3, \ldots$$

is a sequence of countable sets of real numbers, then the set S formed by taking all elements that belong to at least one of the sets S_i is also a countable set.
SEE NOTE 13

2.3.5 Show that if a nonempty set is contained in the range of some sequence of real numbers, then there is a sequence whose range is precisely that set.

2.3.6 In Cantor's proof presented in this section we took for granted material about infinite decimal expansions. This is entirely justified by the theory of sequences studied later. Explain what it is that we need to prove about infinite decimal expansions to be sure that this proof is valid.
SEE NOTE 14

2.3.7 Define a relation on the family of subsets of $\mathbb{R}$ as follows. Say that $A \sim B$, where A and B are subsets of $\mathbb{R}$, if there is a function

$$f : A \to B$$

that is one-to-one and onto. (If $A \sim B$ we would say that A and B are "cardinally equivalent.") Show that this is an *equivalence relation*, that is, show that

(a) $A \sim A$ for any set A.

(b) If $A \sim B$ then $B \sim A$.

(c) If $A \sim B$ and $B \sim C$ then $A \sim C$.

2.3.8 Let A and B be finite sets. Under what conditions are these sets cardinally equivalent (in the language of Exercise 2.3.7)?

2.3.9 Show that an infinite set of real numbers that is countable is cardinally equivalent (in the language of Exercise 2.3.7) to the set $\mathbb{N}$. Give an example of an infinite set that is not cardinally equivalent to $\mathbb{N}$.

2.3.10 We define a real number to be *algebraic* if it is a solution of some polynomial equation

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0,$$

where all the coefficients are integers. Thus $\sqrt{2}$ is algebraic because it is a solution of $x^2 - 2 = 0$. The number π is not algebraic because no such polynomial equation can ever be found (although this is hard to prove). Show that the set of algebraic numbers is countable. A real number that is not algebraic is said to be *transcendental*. For example, it is known that e and π are transcendental. What can you say about the existence of other transcendental numbers?
SEE NOTE 15

2.4 Convergence

The sequence

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \ldots$$

is getting closer and closer to the number 0. We say that this sequence *converges* to 0 or that the limit of the sequence is the number 0. How should this idea be properly defined?

The study of convergent sequences was undertaken and developed in the eighteenth century without any precise definition. The closest one might find to a definition in the early literature would have been something like

> *A sequence $\{s_n\}$ converges to a number L if the terms of the sequence get closer and closer to L.*

Apart from being too vague to be used as anything but a rough guide for the intuition, this is misleading in other respects. What about the sequence

$$.1, .01, .02, .001, .002, .0001, .0002, .00001, .00002, \ldots ?$$

Surely this should converge to 0 but the terms do not get steadily "closer and closer" but back off a bit at each second step. Also, the sequence

$$.1, .11, .111, .1111, .11111, .111111, \ldots$$

is getting "closer and closer" to .2, but we would not say the sequence converges to .2. A smaller number ($1/9$, which it is also getting closer and closer to) is the correct limit. We want not merely "closer and closer" but somehow a notion of "arbitrarily close."

The definition that captured the idea in the best way was given by Augustin Cauchy in the 1820s. He found a formulation that expressed the idea of "arbitrarily close" using inequalities. In this way the notion of limit is defined by a straightforward mathematical statement about inequalities.

Definition 2.6: (Limit of a Sequence) Let $\{s_n\}$ be a sequence of real numbers. We say that $\{s_n\}$ *converges* to a number L and write

$$\lim_{n\to\infty} s_n = L$$

or

$$s_n \to L \text{ as } n \to \infty$$

provided that for every number $\varepsilon > 0$ there is an integer N so that

$$|s_n - L| < \varepsilon$$

whenever $n \geq N$.

A sequence that converges is said to be *convergent*. A sequence that fails to converge is said to *diverge*. We are equally interested in both convergent and divergent sequences.

Note. In the definition the N depends on ε. If ε is particularly small, then N might have to be chosen large. In fact, then N is really a function of ε. Sometimes it is best to emphasize this and write $N(\varepsilon)$ rather than N.

Note, too, that if an N is found, then any larger N would also be able to be used. Thus the definition requires us to find some N but not necessarily the smallest N that would work.

While the definition does not say this, the real force of the definition is that the N can be determined *no matter how small a number* ε *is chosen*. If ε is given as rather large there may be no trouble finding the N value. If you find an N that works for $\varepsilon = .1$ that same N would work for all larger values of ε.

Example 2.7: Let us use the definition to prove that

$$\lim_{n\to\infty} \frac{n^2}{2n^2+1} = \frac{1}{2}.$$

It is by no means clear from the definition how to obtain that the limit is the number $L = \frac{1}{2}$. Indeed the definition is not intended as a method of finding limits. It assigns a precise meaning to the statement about the limit but offers no way of computing that limit. Fortunately most of us remember some calculus devices that can be used to first obtain the limit before attempting a proof of its validity.

$$\lim_{n\to\infty} \frac{n^2}{2n^2+1} = \lim_{n\to\infty} \frac{1}{2+1/n^2} = \frac{1}{\lim_{n\to\infty}(2+1/n^2)}$$
$$= \frac{1}{2+\lim_{n\to\infty}(1/n^2)} = \frac{1}{2}.$$

Indeed this would be a proof that the limit is $1/2$ provided that we could prove the validity of each of these steps. Later on we will prove this and so can avoid the ε, N arguments that we now use.

Let any positive ε be given. We need to find a number N [or $N(\varepsilon)$ if you prefer] so that every term in the sequence on and after the Nth term is closer to $1/2$ than ε, that is, so that

$$\left| \frac{n^2}{2n^2+1} - \frac{1}{2} \right| < \varepsilon$$

for $n = N$, $n = N+1$, $n = N+2$, ... It is easiest to work backward and discover just how large n should be for this. A little work shows that this will happen if

$$\frac{1}{2(2n^2+1)} < \varepsilon$$

or

$$4n^2 + 2 > \frac{1}{\varepsilon}.$$

The smallest n for which this statement is true could be our N. Thus we could use any integer N with

$$N^2 > \frac{1}{4}\left(\frac{1}{\varepsilon} - 2\right).$$

There is no obligation to find the smallest N that works and so, perhaps, the most convenient one here might be a bit larger, say take any integer N larger than

$$N > \frac{1}{2\sqrt{\varepsilon}}.$$

◀

The real lesson of the example, perhaps, is that we wish never to have to use the definition to check any limit computation. The definition offers a rigorous way to develop a theory of limits but an impractical method of computation of limits and a clumsy method of verification. Only rarely do we have to do a computation of this sort to verify a limit.

Uniqueness of Sequence Limits Let us take the first step in developing a theory of limits. This is to ensure that our definition has defined *limit* unambiguously. Is it possible that the definition allows for a sequence to converge to two different limits? If we have established that $s_n \to L$ is it possible that $s_n \to L_1$ for a different number L_1?

Theorem 2.8 (Uniqueness of Limits) *Suppose that*

$$\lim_{n\to\infty} s_n = L_1 \text{ and } \lim_{n\to\infty} s_n = L_2$$

are both true. Then $L_1 = L_2$.

Proof. Let ε be any positive number. Then, by definition, we must be able to find a number N_1 so that

$$|s_n - L_1| < \varepsilon$$

whenever $n \geq N_1$. We must also be able to find a number N_2 so that

$$|s_n - L_2| < \varepsilon$$

whenever $n \geq N_2$. Take m to be the maximum of N_1 and N_2. Then both assertions

$$|s_m - L_1| < \varepsilon \text{ and } |s_m - L_2| < \varepsilon$$

are true.

This allows us to conclude that

$$|L_1 - L_2| \leq |L_1 - s_m| + |s_m - L_2| < 2\varepsilon$$

so that

$$|L_1 - L_2| < 2\varepsilon.$$

But ε can be any positive number whatsoever. This could only be true if $L_1 = L_2$, which is what we wished to show. ■

Exercises

2.4.1 Give a precise ε, N argument to prove that $\lim_{n\to\infty} \frac{1}{n} = 0$.

2.4.2 Give a precise ε, N argument to prove the existence of

$$\lim_{n\to\infty} \frac{2n+3}{3n+4}.$$

2.4.3 Show that a sequence $\{s_n\}$ converges to a limit L if and only if the sequence $\{s_n - L\}$ converges to zero.

2.4.4 Show that a sequence $\{s_n\}$ converges to a limit L if and only if the sequence $\{-s_n\}$ converges to $-L$.

2.4.5 Show that Definition 2.6 is equivalent to the following slight modification:

> We write $\lim_{n\to\infty} s_n = L$ provided that for every positive integer m there is a real number N so that $|s_n - L| < 1/m$ whenever $n \geq N$.

2.4.6 Compute the limit

$$\lim_{n\to\infty} \frac{1+2+3+\cdots+n}{n^2}$$

and verify it by the definition.

SEE NOTE 16

2.4.7 Compute the limit

$$\lim_{n\to\infty} \frac{1^2+2^2+3^2+\cdots+n^2}{n^3}.$$

SEE NOTE 17

2.4.8 Suppose that $\{s_n\}$ is a convergent sequence. Prove that $\lim_{n\to\infty} 2s_n$ exists.

2.4.9 Prove that $\lim_{n\to\infty} n$ does not exist.

2.4.10 Prove that $\lim_{n\to\infty}(-1)^n$ does not exist.

2.4.11 The sequence $s_n = (-1)^n$ does not converge. For what values of $\varepsilon > 0$ is it nonetheless true that there is an integer N so that $|s_n - 1| < \varepsilon$ whenever $n \geq N$? For what values of $\varepsilon > 0$ is it nonetheless true that there is an integer N so that $|s_n - 0| < \varepsilon$ whenever $n \geq N$?

2.4.12 Let $\{s_n\}$ be a sequence that assumes only integer values. Under what conditions can such a sequence converge?

2.4.13 Let $\{s_n\}$ be a sequence and obtain a new sequence (sometimes called the "tail" of the sequence) by writing

$$t_n = s_{M+n} \quad \text{for } n = 1,2,3,\ldots$$

where M is some integer (perhaps large). Show that $\{s_n\}$ converges if and only if $\{t_n\}$ converges.

2.4.14 Show that the statement "$\{s_n\}$ converges to L" is false if and only if there is a positive number c so that the inequality

$$|s_n - L| > c$$

holds for infinitely many values of n.

2.4.15 If $\{s_n\}$ is a sequence of positive numbers converging to 0, show that $\{\sqrt{s_n}\}$ also converges to zero.

2.4.16 If $\{s_n\}$ is a sequence of positive numbers converging to a positive number L, show that $\{\sqrt{s_n}\}$ converges to $\sqrt{L}$.

2.5 Divergence

A sequence that fails to converge is said to *diverge*. Some sequences diverge in a particularly interesting way, and it is worthwhile to have a language for this.

The sequence $s_n = n^2$ diverges because the terms get larger and larger. We are tempted to write

$$n^2 \to \infty \text{ or } \lim_{n\to\infty} n^2 = \infty.$$

This conflicts with our definition of limit and so needs its own definition. We do not say that this sequence "converges to ∞" but rather that it "diverges to ∞."

Definition 2.9: (Divergence to ∞) Let $\{s_n\}$ be a sequence of real numbers. We say that $\{s_n\}$ *diverges* to ∞ and write

$$\lim_{n\to\infty} s_n = \infty$$

or

$$s_n \to \infty \text{ as } n \to \infty$$

provided that for every number M there is an integer N so that

$$s_n \geq M$$

whenever $n \geq N$.

Note. The definition does not announce this, but the force of the definition is that the choice of N is possible *no matter how large M is chosen*. There may be no difficulty in finding an N if the M given is not big.

Example 2.10: Let us prove that

$$\frac{n^2+1}{n+1} \to \infty$$

using the definition. If M is any positive number we need to find some point in the sequence after which all terms exceed M. Thus we need to consider the inequality

$$\frac{n^2+1}{n+1} \geq M.$$

After some arithmetic we see that this is equivalent to

$$n + \frac{1}{n+1} - \frac{n}{n+1} \geq M.$$

Since

$$\frac{n}{n+1} < 1$$

we see that, as long as $n \geq M+1$ this will be true. Thus take any integer $N \geq M+1$ and it will be true that

$$\frac{n^2+1}{n+1} \geq M$$

for all $n \geq N$. (Any larger value of N would work too.) ◀

Exercises

2.5.1 Formulate the definition of a sequence diverging to $-\infty$.

2.5.2 Show, using the definition, that $\lim_{n\to\infty} n^2 = \infty$.

2.5.3 Show, using the definition, that $\lim_{n\to\infty} \frac{n^3+1}{n^2+1} = \infty$.

2.5.4 Prove that if $s_n \to \infty$ then $-s_n \to -\infty$.

2.5.5 Prove that if $s_n \to \infty$ then $(s_n)^2 \to \infty$ also.

2.5.6 Prove that if $x_n \to \infty$ then the sequence $s_n = \frac{x_n}{x_n+1}$ is convergent. Is the converse true?
SEE NOTE 18

2.5.7 Suppose that a sequence $\{s_n\}$ of positive numbers satisfies $\lim_{n\to\infty} s_n = 0$. Show that $\lim_{n\to\infty} 1/s_n = \infty$. Is the converse true?